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An AI-Powered Embodied Virtual Teaching Assistant for Immersive and Interactive Digital Learning

25/04/2026

Faculty of Engineering and Technology

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Document Details

Submission ID

trn:oid:::1:3549870462

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File Name

Pcl_research_paper_final.pdf

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An AI-Powered Embodied Virtual Teaching Assistant for Immersive and Interactive Digital Learning.

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ARTICLE INFO

Keywords:
Artificial Intelligence, Virtual Teaching Assistant, Multimodal Learning, Avatar-Based Systems, Natural Language Processing, Human-Computer Interaction

ABSTRACT

Rapid development in AI technologies has enabled the emergence of different approaches to the creation of intelligent learning environments. Regrettably, many e-learning platforms available today lack the necessary level of interactivity, provide no personalized assistance to the learner, and fail to engage the audience efficiently. This paper outlines the process of designing a virtual tutor or conversational agent capable of interacting via speech recognition and text input and applying AI-based approaches for text, voice, and image generation powered by cloud services. In this regard, the suggested framework depends on the use of AI-based services offered by cloud service providers to ensure multilateral learning through text, voice, and image generation. A prototype of the virtual tutor has been designed and tested for functionality. As shown during testing, the proposed system is capable of processing requests either by voice or text, keeping record of the context, generating responses both by voice and text, as well as creating images according to user requests. To coordinate visual and audio feedbacks, an avatar with an authentic appearance is used. Despite being technologically viable, the discussed approach requires further empirical evaluation and personalization.

1. Introduction

A great deal of transformation has taken place owing to rapid advancement in technologies related to AI in different fields such as health, financial services, and education. Particularly, one field which has gone under dramatic transformation because of internet, use of smart devices, and availability of different types of websites is education [1]. Thanks to the aforementioned technological innovations, now it is possible for students from all backgrounds to easily access learning resources. However, there exist different challenges in these learning platforms.

Education within classrooms provides an effective mode of communication where teachers get to communicate directly with their students. This allows them to analyze the academic performance of students, identify learning deficiencies, and find ways to solve these problems. The interactive approach to learning within classroom education helps students learn more effectively. Contrary to traditional education, many of today's technological systems adopt non-interactive means including recording classes using videos and other methods [9].

This is because the current digital learning environment falls short in terms of context and interaction. The student finds it hard when trying to solve queries or exploring complex topics, since most platforms do not facilitate real-time interaction with intelligent systems.

While some of the systems have integrated a chatbot to assist in providing answers to specific questions, the chatbot is unable to answer any complex question due to its inability to understand a complicated query in its entirety [4]. Therefore, the learners may be provided with vague and irrelevant responses that could affect their educational process in a negative way.

Personalized teaching in online learning environments represents another crucial issue to be considered in the learning process. Individuals have different ways of learning and thinking. They come up with a different level of pre-knowledge. Yet many of the current systems utilize one standard way of distribution of learning materials without considering individuals. Therefore, poor outcomes are experienced by those students who require personalized teaching.

One of such approaches that can address the challenges mentioned above is multimodal learning. Multimodal learning refers to the use of various types of information to improve understanding. In the theory of multimedia learning, multimodal learning is expected to enhance understanding through the use of multiple ways to convey information [9]. Visual information, such as graphs and pictures, allows learners to understand a particular concept easily. The same effect can be achieved through the use of audiovisual information.

However, despite all these advantages, the currently available platforms lack proper utilization of their multimodal properties, leading to an inadequate learning experience. Recent developments in the field of NLP and DL technology made it feasible to create conversational AI that can interact with users just like a human would. This is because of the sophisticated algorithms applied in these conversational agents, such as transformers, which enable them to understand natural languages. So far, these AI agents have demonstrated their efficiency in various fields, from customer service to virtual assistants. In the context of education, conversational AI can open doors for real-time interaction with the content. However, existing uses of conversational AI focus exclusively on text conversation [3].

Interaction within the system constitutes yet another factor that affects the efficiency of the learning process, and here interaction becomes an important element affecting such interaction significantly. Communication of the people is not only exchange of information but also includes the social aspect of interaction. In face-to-face communication within traditional classrooms, the importance of the teacher can play an important role in motivating the learner. In online classes, however, the social aspect is missing, leading to a lack of engagement of the learners in the learning process itself. This has led scholars to search for a solution through the use of avatar and embodied conversational agents [11].

Apart from interaction and engagement, another crucial requirement in designing an educational system in today's context is scalability and performance. This is due to the fact that as the usage of online systems increases, there arises a need for developing systems that will be able to cope with large amounts of information and respond promptly. The cloud computing technology assumes great importance in achieving this purpose as it facilitates scalability and optimal utilization of resources [4]. With the help of cloud computing technology, it is possible to use sophisticated algorithms to make sure that the system works efficiently.

With the development of technology used in the generative AI, new opportunities appear in relation to intelligent education system advancement. On the one hand, the technology enables the generation of realistic text, images, and even speech. This, in turn, gives an opportunity to speak about dynamic educational content, including videos generated with the help of both image generation models and natural language processing [6][3]. On the other hand, the use of an image generation model might prove to be quite useful in illustrating certain concepts. Lastly, with speech synthesis, the technology enables dialogues.

While much progress has been achieved in this field, the adoption of various types of artificial intelligence (AI) in learning activities as an integrated system seems to be quite difficult. Despite the presence of additional functions provided by the modern technology such as the possibility to engage in chatting and the ability to receive audiovisual materials, no integrated system has been developed yet which will incorporate all elements ensuring efficient learning process.

As for addressing the issues mentioned above, it is proposed here to develop the concept of a virtual teacher assistant that would be based on AI and integrate natural language processing, speech recognition, generative AI and avatar generation technologies. It would help the learner to obtain the required feedback from the tutor and use various types of information presentation channels including text, voice, and graphics-based media.

The proposed system architecture is scalable and flexible and, therefore, allows incorporating cloud computing capabilities so as to achieve real-time operations and AI integration effectively [4]. As a result, a large number of users can easily fit into this system without reducing its effectiveness. Besides, the mentioned flexibility opens the way for further development of this system.

The primary objective of the study is to develop a new type of educational system, known as an intelligent education system, which can overcome the gap that exists between the learning systems based on the classroom and those based on computers. With the use of AI and multimodal interaction in the proposed intelligent education system, the interactivity of the system can be improved.

Key contributions made by the current research paper are described below:

- Use of an AI-based embodied virtual teaching assistant for digital learning has been proposed by the researcher.
- An effective multimodal learning environment combining text and verbal conversations with visual content creation using artificial intelligence has been developed to improve the learning process and enhance concept comprehension.
- The proposed system utilizes avatar-based interface that mimics the movements made by the teacher.
- An intelligent response creation method has been devised to create context-aware responses according to the user inputs.
- A cloud-based system architecture has been considered for the implementation of the proposed system for improved performance and scalability.
- A mathematical model for functioning of the proposed system has been designed which illustrates how responses are generated and learning takes place.
- Functioning of the proposed system has been demonstrated using a prototype and testing.
- The proposed system has been compared with numerous other conventional systems including chatbots and e-learning systems.

2. The Related Work

The development of Artificial Intelligence at a very fast pace has contributed significantly to the creation of intelligent education systems. Advanced intelligent education systems are developed to create a greater level of accessibility, automation, and interaction through technology [10].

Rapid developments in Artificial Intelligence have resulted in advancements in learning systems that can be categorized into chatbot learning systems, intelligent tutoring systems, and multimodal learning systems. The main reason behind the development of these learning systems is their ability to increase access and automate knowledge and interact with learners in an effective manner [10]. However, there are many learning systems which do not possess multimodal, contextual, and humanlike interaction capabilities.

2.1 Intelligent Tutoring Systems

Intelligent Tutoring System (ITS) is one of the application areas where the use of AI can be found in education domains for providing personalizing education in relation to the performance of the individual. This type of system operates like a one-to-one tutor, analyzing the answers given by the student and giving suggestions accordingly [2]. The learning process in the ITS system can be described as follows:
The mathematical formulation of adaptation for learning is:

$$Kt+1 = Kt + \Delta K$$

Here Kt represents the present knowledge state,
and ΔK represents the increment in knowledge state.
The main limitation of traditional ITS is that they have been confined with certain set of predefined rules, and as a result they cannot accommodate open-ended problems. [10].

2.2 Conversational AI and Natural Language Processing

With the progress made in NLP, there is now the possibility of building conversational AI that can interact with users in a manner similar to how humans interact using approaches such as tokenization, semantic understanding, and context embedding [4].
Through recent advancements in transformer-based models, the creation of responses based on contextual information is now possible. The formulation for generating response probabilities is expressed as:

$$P(R \mid Q) = \prod_{i=1}^{n} P\,(r_i \mid r_1, r_2, \ldots, r_{i-1}, Q)$$

where Q denotes the query input, and R is the sequence of responses produced.
Even though these artificial intelligence systems are interactive in character, they still have many limitations that include limiting the transfer of data to text communication only.

2.3 Multimedia and Multimodal Learning Systems

In a multimedia learning environment, multiple modes of information transmission are employed to facilitate learning processes. According to multimedia learning theory, the use of various stimuli modes improves cognition and learning processes [9]. The effectiveness of multimedia learning is expressed through the equation below:

$$L = f\,(T, A, V)$$

where T, A, and V denote input from texts, sounds, and images respectively

Although possessing many positive features, most of these systems cannot facilitate real-time interaction or dynamic content creation [20].

2.4 Avatar-Based and Embodied Learning Systems

Avatar-based systems use human-like interfaces in digital education environments, enhancing engagement and interactions. Avatar-based system mimics actual teaching actions by using a virtual entity that is capable of communicating both verbally and non-verbally [11].
The effectiveness of interaction in an avatar-based system can be expressed as:

$$E = \alpha C + \beta I$$

where C denotes the quality of communication and I represents the level of interaction.
The presence of these pedagogical agents has been shown to significantly increase the motivational and interactive components of learning, a phenomenon called the "persona effect" [19]. Sadly, modern avatar systems do not make use of intelligent AI technology.

2.5 Generative AI in Educational Systems

With the advent of AI technology, there have been several technological advancements that have led to the creation of dynamic content. These include techniques like GANs and transformers, and the process of generating content creates educational content that is highly realistic [6], [3].
Content generation involves:

$$X_{gen} = G(z)$$

where z refers to the hidden variable, and G refers to the generation function. These methods may be used in the creation of visual and interactive material, which would promote better understanding. However, their use in educational software systems remains limited.

2.6 Speech Processing and Voice-Based Interaction

Speech Recognition and Synthesis make it easy for users to interact with computers by speech. It is possible to use technologies of speech recognition and synthesis to translate a speech signal into text or vice versa [7]. The following picture illustrates translation of speech into text:

$$T = f(S)$$

Where S represents the speech signal and T refers to the text output.

Although voice technology increases accessibility, it does not always become part of a complete learning system, but rather works as a stand-alone component.

2.7 Limitations of Existing Systems

However, there are a few limitations associated with the existing methods. Chatbot methodology uses interaction without multimodality, multimedia methodology offers better presentations but does not provide any interactivity, while the avatar methodology offers interaction without intelligence and context. This demonstrates that there is a requirement for an integrated system that combines these three features together.

2.8 Summary and Research Gap

To conclude, the above-mentioned research studies have been very effective in studying different components in digital education such as tutors, intelligent agents, and multimodal learning. Undoubtedly, at present, there is no existence of any system where all these components will be merged together.
The proposed system is developed based on natural language processing, speech interface, generative artificial intelligence, and avatar interface technologies. It has been developed to address any problems faced by the previous studies [1], [4], [9].

3. The Proposed System / Algorithm

The Proposed System

In this paper, we will attempt to improve the capabilities of the digital learning system by highlighting the limitations of current solutions that utilize chatbots and e-learning programs, such as low interactivity, context understanding, and multimodality. In order to solve the above problems, we will offer a solution that will leverage AI technology to develop an embodied virtual assistant for education.

Our proposed solution will feature several modules of artificial intelligence for adaptive learning. In addition to traditional methods used in other solutions, such as the employment of a single communication channel, our solution will incorporate multimodal communication to maximize the learning process. Our solution is designed to respond adequately to context, explain concepts visually, and mimic human behavior in education through the use of avatars. The overall purpose of our solution is to enhance the learning process and ensure real-time interaction. It can be defined as an optimization task:

$$L_{opt} = \max\,(E + I + A)$$

where E represents engagement, I denotes interactivity, and A indicates adaptability of the system.

3.1 System Modelling and Purpose of Equations

The suggested architecture is to be represented as an interactive system in which the process of processing user inputs consists of several steps, including data collection, natural language processing, answer creation, and output generation.
User input may be viewed as follows:

$$Q = \{q1, q2, \ldots, qn\}$$

Input processing by the proposed system entails employing an NLP model that produces a corresponding response R:

$$R = f(Q, C)$$

where C refers to the contextual data acquired from previous user conversations.
For proper processing and generation of relevant outputs, the following probability function is used for optimization:

$$P(R \mid Q) = \prod_{i=1}^{n} P(ri \mid Q, r1, r2, \ldots, ri-1)$$

3.2 State-Based Interaction Modelling

The process of interaction between user and the system is described by means of state transitions, wherein every interaction becomes a state transition from one state to another.
Let the state of the system be described as:

$$St = (Qt, Ct),$$

where Qt is the query issued by the user at time t and Ct is the context at that time. Next state can be written as:

$$St+1 = St + \Delta S,$$

where ΔS is the change in context and knowledge. In order to enhance interactions, system uses feedback from the user along with past interactions.

3.3 Multimodal Response Generation

One of the most important attributes of the new system under consideration is multimodal response generation, where responses are created using text, speech, and visuals.

Overall responses can be expressed using the formula:

$$O = T + V + A$$

where T refers to text response, V stands for visual content (pictures), and A is an audio response.

3.4 Adaptive Learning Mechanism

To enhance personalization, there is a learning adaptation process that learns how to respond based on interaction.
The formula for this learning adaptation will be:

$$Ut+1 = Ut + \alpha(Ft)$$

where Ut is the knowledge of the user, Ft is the feedback, and α is the learning rate.
This ensures the growth of the system and provides a better response to the users.

3.5 System Workflow and Architecture

The proposed design follows a multi-layer intelligent architecture that consists of several different types of AI technologies that help deliver an interactive and engaging learning experience. As shown in Fig. 1, the proposed solution has the following layers:
User Interaction Layer, Input Processing Layer, AI Intelligence Layer, Multimodal Generation Layer, Avatar Interaction Layer, and Cloud Computing Layer.

The user interaction layer forms the user interface that helps the user interact with the system and submit their input data through both text and voice input modes. The input processing layer helps convert the voice input into text while carrying out various other pre-processing activities like tokenization and normalization.

The intelligence layer forms the core of any AI-powered solution, as it is at this level that natural language processing is carried out in order to comprehend the query submitted by the user and provide a corresponding response. Context management helps the solution remember previous conversations so that they may be taken into account while generating a response.

The multimodal generation layer enhances the learning experience since the generated response can come in several different formats like texts, images, and voice responses. This is achieved using various generative AI models and text-to-speech models.

The avatar interaction layer embodies the human face in the AI-supported system, whereby the responses are generated by virtual human tutors. The lip-syncing and facial expressions of the avatars determine how realistic the behaviors appear.

The output delivery layer marks the stage where the response to the user's question is delivered using a combination of texts, audio, and visual components. Furthermore, the cloud and data layer guarantees the scalability and efficient data management in the system by collecting data on user behavior and combining AI services.

Such an architecture enables real-time query processing, scalability, and integration of different AI services.

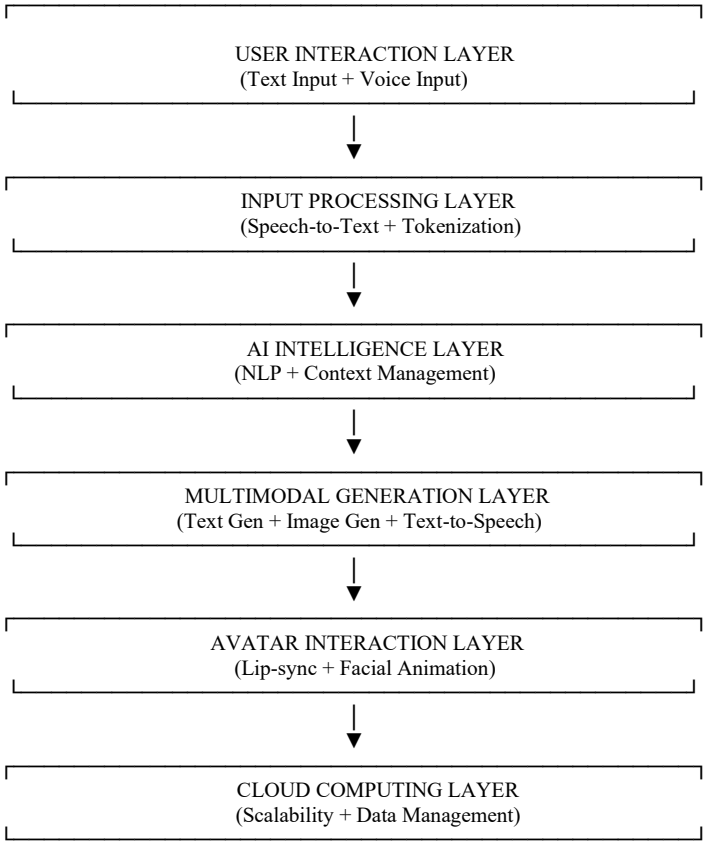

Fig 1: Architecture of the Proposed AI-Based Virtual Teaching System

3.6 Time Complexity Analysis

Time complexity is one factor that will be used to determine the effectiveness of this system design. Here, N represents the number of input data from users, while M represents the complexity of the AI model.
This means that the time complexity equation can be written as follows:

$$O\ (N \times M)$$

Cloud computing allows operations to take place within a negligible amount of time.

3.7 Summary of Proposed System

To sum up, the proposed system has incorporated several forms of AI in a single system, making it possible for learning to be enjoyable and interactive. This is possible due to the incorporation of Natural Language Processing, voice recognition, Generative AI, and avatars in the proposed system, making it possible for the proposed system to overcome limitations found in today's learning systems.
These include modeling, adaptation, and multiple channels.

4. Results and Demonstration

4.1 Prototype Implementation

The basic implementation of the AI-Based Virtual Teaching Assistant prototype was performed successfully. The basic functionality which has been included into the system includes:

- Audio recognition as one of the input modes;
- The ability to understand natural language used in questions of the users;
- Creation of responses using the large language model;
- Converting text into voice to deliver responses in spoken format;
- Image generation to create images that explain the response;
- Conversational context handling;
- Avatar interface.

The app runs online and allows the user to communicate with the app in the form of speech or text by asking a question and receiving an answer via multiple interfaces such as audio, text, or even images. The avatar itself moves just lips during the dialogue.

4.2 Functional Testing and Observed Behavior

The prototype underwent testing through scenarios comprising a series of 20 educational questions concerning various scientific facts, mathematics, or computing concepts.

Behavior Observed:

When faced with simple factual questions such as "What is the capital of France?" or "Define photosynthesis?" the system produced the correct answer in text form as well as spoke the same aloud. However, in case of voice recognition, the prototype did not respond to questions containing ambient sounds.

Conceptual queries requiring an explanation led to satisfactory explanations in paragraph form and even context in case of additional questions such as "give me an example" or "can you explain that more clearly?"

Queries requiring a visual representation led to the successful creation of diagrams representing the concepts asked about. However, once again, the efficiency of this process depended on how difficult the concept was.

The system included a rudimentary avatar that provided simultaneous speech to the text displayed; however, its behavior remained unchanged regardless of the type of question posed.

4.3 Identified Limitations

Here are some of the limitations noticed in the functional testing process:

As a whole, the entire system depends extensively on calling API services for recognizing speech, creating a language model, and generating images. Consequently, the response time varied significantly based on internet connectivity speed and load on the API services.

Sometimes the generated image had nothing to do with the user's input, particularly in the case of abstract or unique words. For example, if the user entered "diagram of backpropagation algorithm," the image generated was of some random mathematical formulas.

The lip sync of the virtual assistant did not always match with the generation of speech, which made the process unnatural. Furthermore, the virtual assistant does not have the capability to express emotions through facial expressions based on the context of the conversation.

No functionality is available for creating persistent user profiles and adjusting to their learning experience during the sessions. Every new session starts with zero knowledge about the user's previous experiences and accomplishments.

If a question requires intricate reasoning or solving a problem through multiple steps, the answer might not be entirely accurate or complete.

4.4 Successful Demonstration Cases

Still, the prototype has managed to convince that there are several aspects that distinguish it from classic e-learning applications:

For example, in one case, after a user had input "What is machine learning?" and then, immediately after, said "Give me an example," the system understood that the latter query was concerned with machine learning and provided an example associated with spam filter technologies. It demonstrates the system's capability to retain context across two consecutive interactions.

Also, in another case, after a user had instructed the system "Draw a binary search tree," a corresponding graphical depiction of nodes appeared as a result of interaction. Though it was not as accurate as required, nonetheless, it communicated the necessary meaning to the user.

Lastly, in the case of a voice test, after a user had issued a command "Define quantum entanglement," a voice recognition process occurred, followed by an interpretation of the query, providing a textual response, which was also read out and displayed visually.

4.5 Summary of Findings

Such an application can be used as a proof of concept that proves that a number of AI-based technologies can be used together to create an AI tutor (speech recognition, speech synthesis, natural language processing and understanding, natural language generation, and avatar animation). The separate components work very effectively in isolated tests, but the overall sequence works less effectively compared to each separate component but creates an adequate response to the user's query.

However, it cannot be considered a working prototype since it requires some modifications to provide faster responses, generate images, create avatars, and offer personalized responses to the user's questions.

4.6 Future Work Directions

With respect to the status and limitations of the existing prototype, the following is anticipated in the future work:

Short Term (over next 2-3 months):

- Conduct user testing involving 10-15 students to gather their feedback concerning usability and relevance
- Implements response caching system for questions that are often asked in order to reduce lag time
- Lip-syncing capability of the avatar through browser-based Web Speech API

Medium Term (over next 3-6 months):

- Trying out the product in real-world settings such as in the class environment for one week
- Logging and analysis of logs to determine common failure points
- Basic personalized features such as storing user names and preferred topic of discussion

Long Term (over next 6-12 months):

- Substitute all cloud-based APIs with local implementation of opensource models such as Whisper (speech), Llama 3 (texts) and Stable Diffusion (images) in order to get rid of any external dependence
- Evaluate the effectiveness of the product versus the traditional chatbot in a laboratory-like setting for competitive edge in learning and interaction
- Use voice emotion detection technology to modify teaching style

4.7 Conclusion

The aforementioned research paper elaborated on the architecture of an AI-powered virtual assistant for teaching applications that make use of the potential of speech recognition, natural language processing, text-to-speech conversion, image generation, and avatar modeling. A working model of the system was constructed demonstrating the technical feasibility of the proposed approach..

Moreover, it was evaluated concerning its functionality and proven to have the ability to recognize the voice/text commands, keep a track record of the conversation, provide corresponding responses, convert them into a speech form, and create pictures, where needed. Avatar illustrates the presence, which operates along with speech synthesis.

However, there are a lot of limitations, such as dependency on APIs, inability to create pictures, basic avatar animations, lack of personalization capabilities, and, finally, no users' evaluations were conducted yet. Consequently, this system can only be used as a proof of concept, but not as an educational software product.

Nonetheless, successful integration of different Artificial Intelligence techniques into one system aimed at providing assistance to teachers should be regarded as a step toward enhancing interaction and engagement in e-learning. In the nearest future, work will be done on testing, modeling, avatar creation, and customization capabilities implementation. Taking these steps into consideration, the proposed solution possesses all necessary prerequisites for bringing closer two distinctively different approaches to learning

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